

WAYFINDER: A Real-Time GPS Based Bus Tracking and Smart Bus Pass Verification System

UI/UX Design Document

Project-Based Placement Readiness Programme
Training and Placement Cell

Team Number: XI | Team Name/Repository: Team_WAYFINDER

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Abstract

This document presents the User Interface (UI) and User Experience (UX) design for the WAYFINDER system. It describes the design goals, the three distinct user roles the interface serves, the navigation structure for each role, and a screen-by-screen breakdown of the mobile and web interfaces designed for the project. The designs were created as high-fidelity prototypes to establish a clear, concrete reference for the functional and non-functional requirements defined in Phase 2, and will guide the implementation phase.

Keywords—UI/UX Design, Prototyping, Mobile Application Design, Admin Dashboard, Design System

1. Introduction

The purpose of this document is to communicate how users will interact with the WAYFINDER system across its three distinct interfaces: the Student/Staff mobile application, the Driver mobile application, and the Transit Admin web dashboard. Each interface was designed as a high-fidelity prototype to give the team, and this documentation, a concrete visual reference of the proposed system ahead of development, in line with the objective of Phase 3 – Design & Planning.

The following sections outline the design goals and principles guiding the interface, the roles the system serves, the navigation structure of each application, and a detailed walkthrough of every designed screen with supporting figures.

2. Design Goals & Principles

- Clarity at a glance: status information (bus delays, pass validity, alerts) is communicated through color and short labels before any detail is read.
- Minimal user input: core actions – starting tracking, checking a pass, viewing an ETA – are reachable within one tap from the home screen of each app.
- Consistency across roles: the Student/Staff and Driver mobile apps share the same bottom navigation pattern and visual language, so switching between them feels familiar.

- Real-time feedback: live status (tracking active, bus approaching, delay alerts) is always visible rather than requiring a manual refresh.
- Role-appropriate density: the Admin dashboard favors information density (tables, live feeds) suited to desktop use, while the mobile apps favor a simplified, single-focus layout per screen.

3. User Roles

Role	Platform	Primary Goals
Student / Staff	Mobile app	Track live bus location and ETA, receive arrival alerts, view digital bus pass.
Driver	Mobile app	Start/stop GPS tracking with minimal interaction, raise SOS/emergency alerts.
Transit Admin	Web dashboard	Monitor fleet and routes in real time, manage dispatch status, issue and review bus passes.

4. Navigation Structure

The Student/Staff and Driver mobile apps share a bottom navigation bar with three destinations – Map, Pass, and Alerts – keeping the core tasks reachable from anywhere in the app. The Driver app reduces this further to a single-purpose tracking screen, since a driver's primary task is starting or stopping tracking rather than browsing routes.

The Transit Admin web dashboard uses a persistent left sidebar (Fleet, Dispatch, Passes) instead of bottom navigation, matching conventional desktop admin-panel layout and allowing quick switching between fleet-wide views.

5. Screen-by-Screen Design

5.1 Student / Staff Application

5.1.1 Live Map & Route Tracing – The home screen shows the selected route on a live map with the bus's current position and the user's location. A status banner surfaces immediate events (e.g. “Bus Departed”) with a one-tap action to track the next bus. Below the map, a persistent Live ETA card shows minutes-to-arrival and an on-time/delayed

indicator, with a toggle to enable a 10-minute-prior arrival alert.

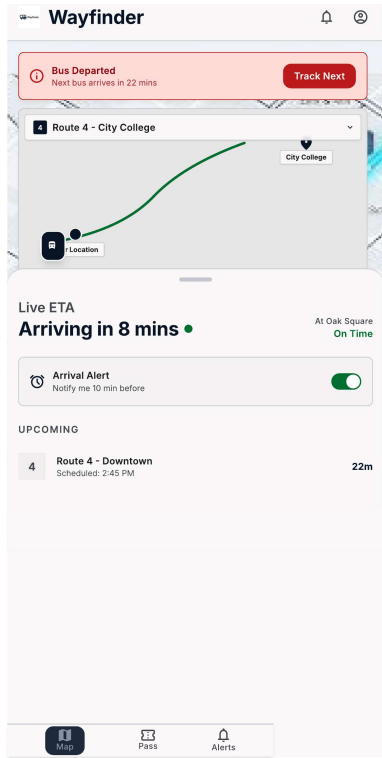


Fig. 1. Live Map & Route Tracing – Student/Staff app.

5.1.2 Digital Bus Pass – The pass screen presents a single card styled like a physical ID, colored according to validity status (green for valid). It shows the student's photo, name, department, student ID, route access, and expiry date, along with a share action for presenting the pass digitally to conductors or staff.

5.1.3 Route Alerts & Notifications – A chronological feed of route-related events, each tagged with a severity label – CRITICAL (service disruption), DELAYED (traffic-related delay), or TRACKING (informational, e.g. bus approaching) – so a commuter can judge urgency without reading the full message.

5.2 Driver Application

5.2.1 Driver Dashboard – A deliberately minimal single screen showing the assigned bus and route, an online/offline status pill, a large Start Tracking control, and a persistent geofence notice explaining that tracking will stop automatically on reaching the college campus. A separate, visually distinct SOS/Emergency control is always available for breakdown or accident alerts.

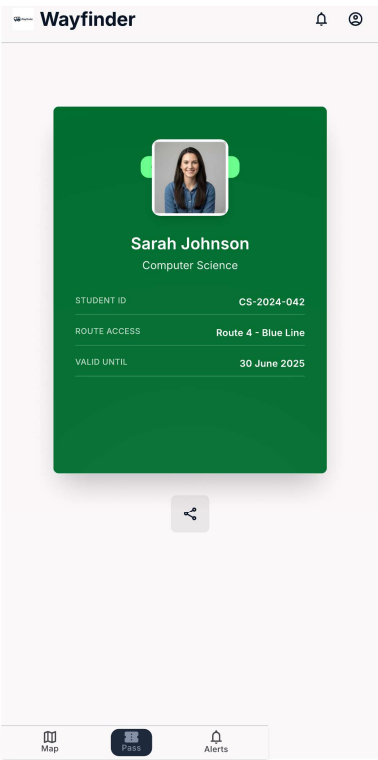


Fig. 2. Digital Bus Pass – Student/Staff app.

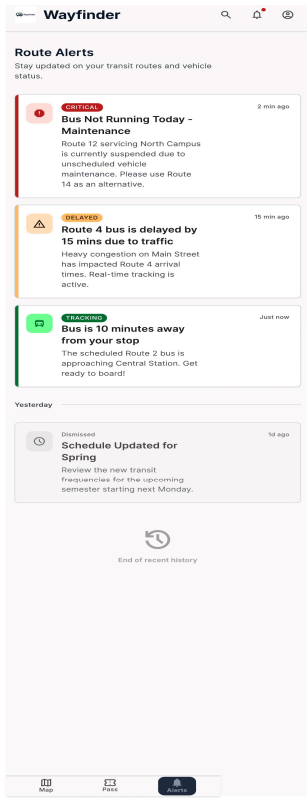


Fig. 3. Route Alerts & Notifications – Student/Staff app.

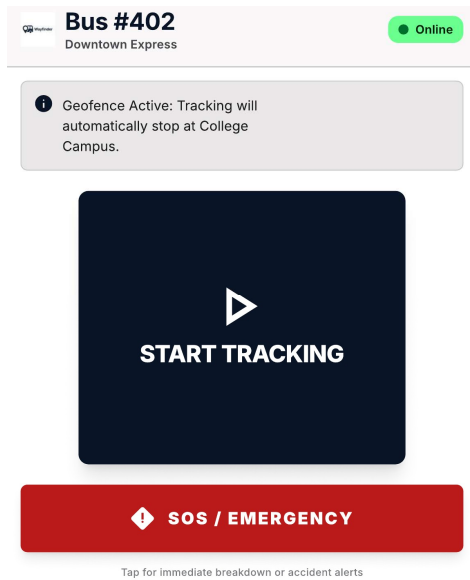


Fig. 4. Driver Dashboard – Driver app.

5.3 Transit Admin Web Dashboard

The Admin dashboard is designed for desktop use by transit authority staff and is shown across the full page width below to preserve the detail of its tables and live panels.

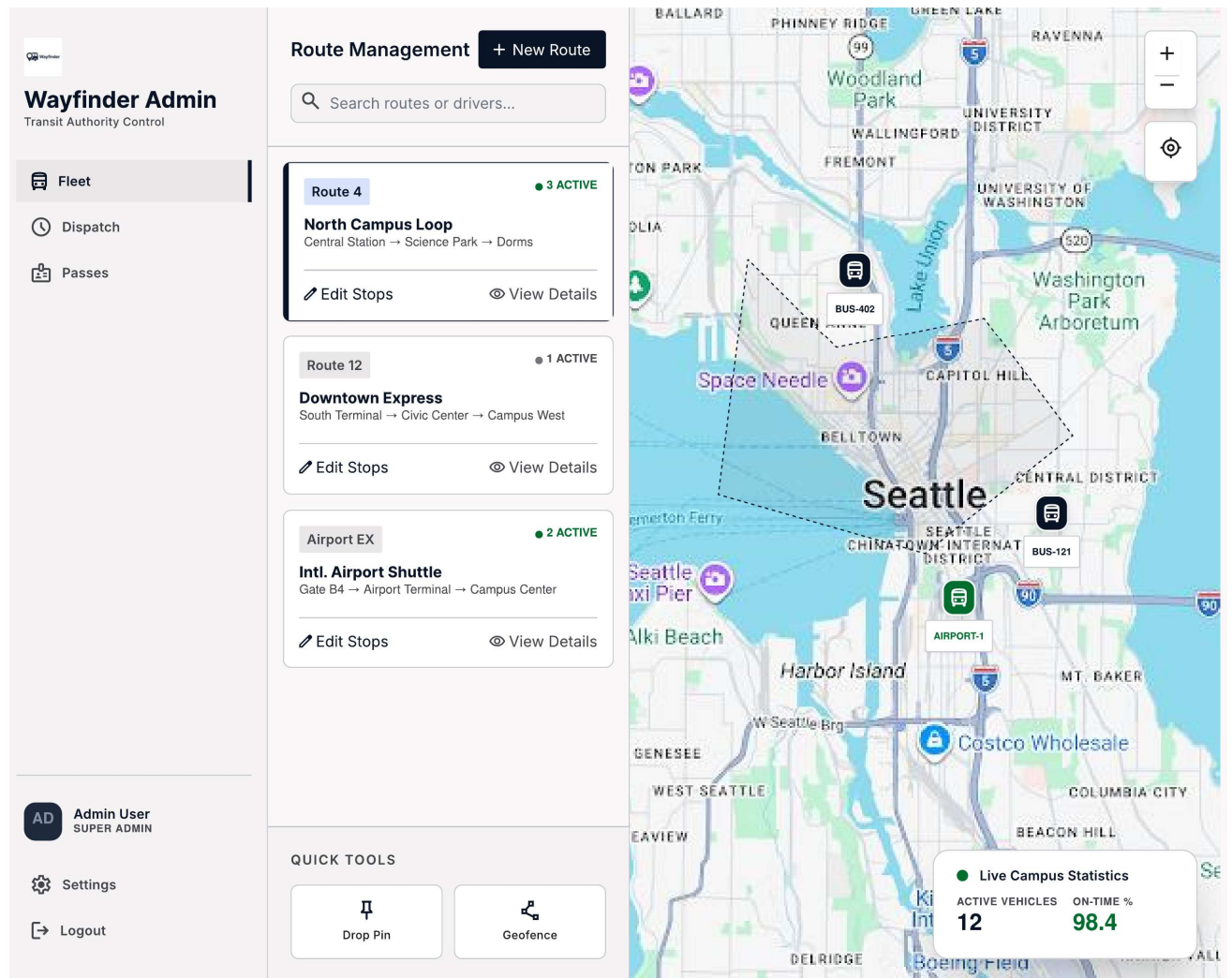


Fig. 5. Fleet Overview & Route Management – Transit Admin dashboard.

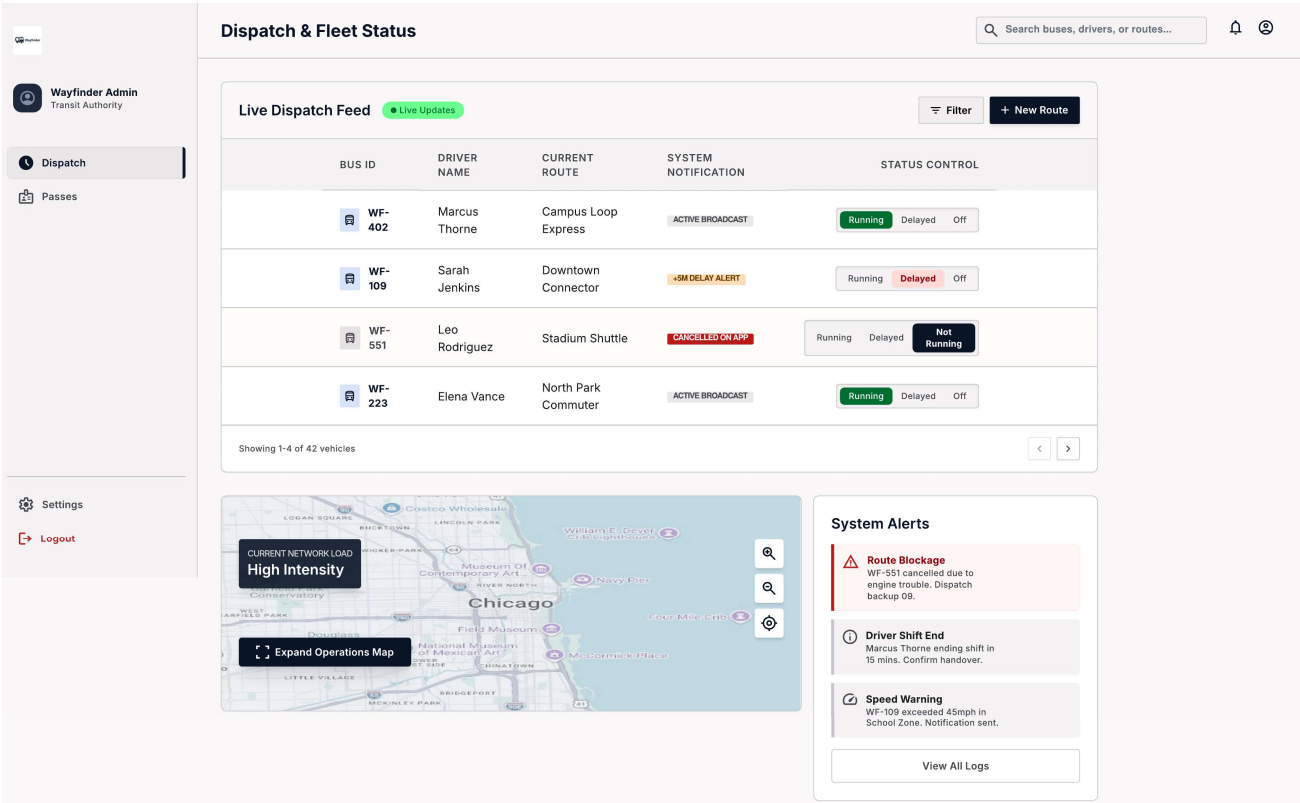


Fig. 6. Dispatch & Fleet Control – Transit Admin dashboard.

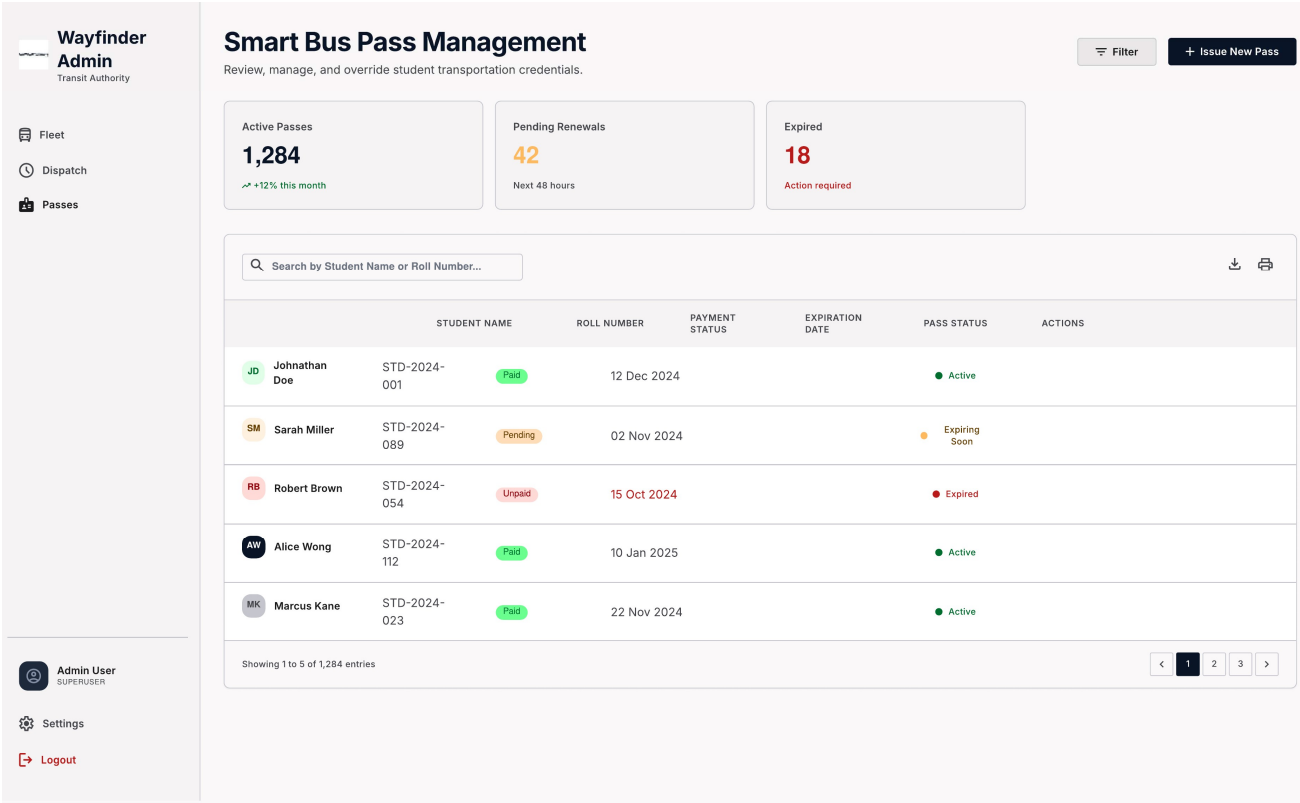


Fig. 7. Smart Bus Pass Management – Transit Admin dashboard.

6. Visual Design System

A consistent color system is used to communicate status across all three interfaces without relying on text alone:

Color	Meaning
Green	Valid / Active / Running / On time
Amber / Orange	Delayed / Pending / Expiring soon
Red	Critical / Expired / Cancelled / Emergency
Dark navy	Primary actions and admin sidebar (brand anchor)

Typography uses a clean sans-serif throughout for on-screen legibility, with a consistent card-and-pill component pattern (rounded status badges, bordered cards with a colored left accent for severity) reused across the alerts feed, dispatch feed, and pass status indicators so that a user only has to learn the pattern once.

7. Mapping to Functional Requirements

Each screen was designed to directly satisfy the functional requirements defined in 03_Requirement_Analysis_and_Literature_Survey.pdf. Live GPS tracking and per-stop ETA are addressed by the Live Map & Route Tracing and Driver Dashboard screens; push notifications 10 minutes prior to arrival are addressed by the Arrival Alert toggle and Route Alerts feed; digital bus pass validation is addressed by the Digital Bus Pass and Smart Bus Pass Management screens; and fleet-wide monitoring and dispatch control are addressed by the Fleet Overview and Dispatch & Fleet Control screens.

8. Conclusion

The designs presented in this document translate the WAYFINDER proposal and Phase 2 requirements into a concrete, screen-level reference covering all three user roles. They establish the navigation structure, visual language, and status-communication system that development will follow in the coming phases, and will continue to be refined as implementation progresses.